Industrial Dissertation - NUIST

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Industrial Dissertation - NUIST (A35627)

Short Title:	Industrial Dissertation (NUIST)		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Creativity and Innovation		
Author	CLAIRE.FITZPATRICK		
Credits	30	**Level:**	Postgraduate

Description of Module / Aims

The Industrial Project is a very significant component of the programme, providing students with the opportunity to apply engineering and related principles and theory in an innovative manner, or to develop innovative solutions to challenging engineering and scientific problems. Ultimately, the project component enables participants to `individualise' the award. Students are required to submit a written project proposal and to give an oral presentation of the proposal. The proposal is required to incorporate the context of the project in research and industrial arenas. A substantial research element is required, and the formulation of specific and measurable objectives are an inherent part of the process, together with an appropriate schedule of activities in order to ensure the achievement of deliverables. Students are required to develop a `solution' for their project, or substantially progress the innovative application of the relevant technology / technologies. A strong element of application, integration, implementation, and analyses of the relevant underlying engineering and scientific principles is an essential part of this module. Students are required to write a dissertation, and to make final oral and poster presentations of their work. Full details are provided in the document `Project and Dissertation Guidelines' which is reviewed and updated annually.

Programmes

DISS-0069	MEng in Electronic Engineering (WD_EELEN_R)	1 / 3 / M
	MEng in Electronic Information Systems (WD_EELEN_RI)	2 / 3 / E
	MEng in Electronic Information Systems (WD_EELEN_RI)	2 / 4 / E
	MEng in Electronic Information Systems (WD_EELEN_RI)	3 / 5 / E
	MEng in Electronic Information Systems (WD_EELEN_RI)	3 / 6 / E
DISS-0069	MSc in Innovative Technology Engineering (WD_CINNT_R)	1 / 3 / M

Indicative Content

- Formulation of project proposal, together with scheduling activities
- Review of literature, and 'state-of-the-art'
- Presentations, team-based discussions and other communications.
- Project management activities.
- Report writing.
- Evaluation, analyses and discussion of results.
- Ethical issues in projects.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify, formulate, analyse and solve engineering problems in the context of an industrially-relevant project.
2. Formulate a coherent project proposal, including appropriately identified objectives, methodology, milestones and deliverables.
3. Design a system, component, process or material(s) to fulfil specified industrial requirements.
4. Derive and apply solutions from a knowledge of sciences, engineering sciences, technology and mathematics.
5. Design and conduct experiments, and to analyse and interpret data in accordance with good scientific and engineering practice.
6. Evaluate where the project work fits into ongoing related work by other academics and practitioners.
7. Identify and evaluate the impact of ethical issues where appropriate, and to embrace the responsibilities of the engineering profession towards people and the environment.
8. Work effectively as an individual, in teams and in multidisciplinary settings together with the capacity to continue to improve their own knowledge and expertise.
9. Select and utilise appropriate tools to effectively communicate their work in an academic, industrial and commercial setting, and to society in general.

Learning and Teaching Methods

- Presentations.
- Project-based activities, specific to the nature of the project.
- Student project log, with supervisor feedback.
- Project participation.
- Report writing.
- Presentation of findings.
- Final report of findings.

Assessment Methods

	Weighing	Outcomes Assessed
Final Project	65%	1,3,4,5,6,7,8,9
Continuous Assessment	35%	
Assignment	10%	1,2,6,7,9
Presentation	5%	1,2,6,7,9
Supervisor's Report	3%	1,2,6,7,9
Supervisor's Report	7%	1,3,4,5,6,7,8,9
Presentation	5%	1,2,3,4,5,6,7,8,9
Presentation	5%	1,2,3,4,5,6,7,8,9

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

40% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Independent Learning	786	